Predicting Heart Disease Using ANN

A PROJECT REPORT
For Introduction to AI – (AI101B) Session (2024-25)

Submitted By

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CERTIFICATE

Certified that **Subhash Kumar 202410116100211, Shubham Singh 202410116100204,** has/ have carried out the project work having "**Predicting Heart Disease Using ANN**" (**Introduction to AI, AI101B**) for **Master of Computer Application** from Dr. A.P.J. Abdul Kalam Technical University (AKTU) (formerly UPTU), Lucknow under my supervision. The project report embodies original work, and studies are carried out by the student himself/herself and the contents of the project report do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.

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ABSTRACT

Heart disease remains one of the leading causes of mortality worldwide, making early and accurate detection crucial for improving patient outcomes and reducing healthcare burdens. This project presents a predictive model for heart disease diagnosis using Artificial Neural Networks (ANN), a powerful deep learning technique capable of capturing complex, nonlinear relationships within medical data. The model is trained on the widely used UCI Heart Disease dataset, which includes key clinical features such as age, blood pressure, cholesterol levels, and chest pain type.

The data undergoes rigorous preprocessing, including normalization, to ensure effective learning. The ANN architecture consists of multiple dense layers with ReLU activation functions, batch normalization for stability, and dropout layers to prevent overfitting. The final output layer uses a sigmoid activation function for binary classification. The model is optimized using the Nadam optimizer and trained with early stopping to avoid overfitting.

Performance is evaluated using key metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix. Additionally, SHAP (SHapley Additive exPlanations) is employed to interpret the model's predictions and highlight the most influential features, enhancing transparency and trust.

The results demonstrate that the ANN model achieves high predictive accuracy and reliability, highlighting the potential of deep learning in medical diagnostics and supporting its integration into real-world clinical decision support system

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INTRODUCTION

1.1 PROJECT DESCRIPTION

Heart disease is a critical public health concern and one of the leading causes of death globally. According to the World Health Organization (WHO), millions of lives are lost each year due to cardiovascular diseases, many of which could be prevented through early diagnosis and intervention. Traditional methods of diagnosis, including ECGs, angiograms, and physical symptoms analysis, often require considerable time, clinical expertise, and expensive procedures. Moreover, manual diagnosis is susceptible to human error and inconsistency.

With the rapid advancement of **Artificial Intelligence (AI**), especially in the field of machine learning and deep learning, predictive models are now being explored and adopted to support faster and more accurate disease diagnosis. This project aims to develop a machine learningbased classification model using Artificial Neural Networks (ANNs) to predict the likelihood of heart disease in patients based on various clinical attributes.

The dataset used in this project is the **UCI Heart Disease dataset**, which includes key patient information such as age, sex, chest pain type, resting blood pressure, cholesterol levels, fasting blood sugar, ECG results, maximum heart rate, exercise-induced angina, ST depression, and others. These features are crucial indicators for diagnosing the presence of heart disease.

The core objective of this project is to build a robust ANN model capable of learning from historical data and accurately predicting whether a person is likely to have heart disease. To achieve this, the data undergoes preprocessing techniques such as normalization and splitting into training and testing sets. The ANN is then constructed with multiple hidden layers, ReLU activation functions, dropout for regularization, and batch normalization for training stability. The final layer uses a sigmoid function for binary classification.

Model performance is evaluated using various metrics, including accuracy, precision, recall, F1score, ROC-AUC, and confusion matrix. Furthermore, SHAP (SHapley Additive exPlanations) is applied to interpret the model's predictions and identify the most influential features, ensuring transparency and interpretability — critical aspects in healthcare applications.

This project demonstrates the practical application of deep learning in predictive medicine and highlights the potential of ANN models as supportive tools for healthcare professionals in early and efficient heart disease diagnosis.

Key Objectives:

- Develop Accurate **ANN Model** Build an Artificial Neural Network (ANN) model that can classify patients as either having heart disease or not, based on clinical attributes from the UCI Heart Disease dataset.
- Implement Effective Data Preprocessing Normalize and prepare the dataset to ensure that all input features are scaled appropriately, improving the convergence and performance of the ANN during training.
- Optimize Model Architecture for Generalization Design the ANN using multiple hidden layers, ReLU activation functions, batch normalization for training stability, and dropout layers to prevent overfitting.
- Use **Early Stopping** to Prevent Overtraining Apply early stopping during model training to halt training when the validation loss no longer improves, thereby reducing the risk of overfitting.
- Evaluate Using **Multiple Performance Metrics** Assess the model's effectiveness using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix for a comprehensive evaluation.
- Enhance Model Interpretability with **SHAP**, Use SHAP (SHapley Additive exPlanations) to understand which features most influence the model’s predictions, increasing transparency and trust in a medical setting.

1.2 PROJECT SCOPE

The scope of this project focuses on the design, development, and evaluation of a deep learningbased predictive system for heart disease diagnosis. By leveraging Artificial Neural Networks (ANNs), the project aims not only to achieve high accuracy in prediction but also to ensure transparency, scalability, and real-world applicability in clinical environments. The scope is broadly categorized into the following key areas

1. Apply Deep Learning to Medical Data Classification

The core aim is to utilize deep learning techniques to classify whether a patient is at risk of heart disease based on clinical data. This includes:

- Building a binary classification model using an **Artificial Neural Network (ANN)**.
- Training the model on the UCI Heart Disease dataset, which consists of key patient attributes such as age, blood pressure, cholesterol, chest pain type, etc.
- Utilizing ReLU activation, dense layers, and sigmoid output to perform effective classification.
- Demonstrating how deep learning can enhance traditional diagnostic approaches in healthcare by offering speed, scalability, and data-driven decision-making.

2. Ensure Transparency in Predictions via SHAP Analysis

Transparency and explainability are critical for the adoption of AI in medical fields. This project addresses this through:

- Integrating **SHAP (SHapley Additive exPlanations)** to interpret the ANN's predictions.
- Explaining individual model predictions by quantifying the contribution of each input feature (e.g., age, cholesterol, chest pain type) to the final decision.
- Generating SHAP summary plots to visualize the overall impact of each feature on model outcomes.
- Supporting clinicians and stakeholders in understanding how the AI arrives at a decision, fostering trust in the system.

3. Improve Training Efficiency and Generalization Through Regularization Techniques

To ensure the model is not just accurate but also efficient and generalizable across new data, the project includes:

- Implementing **Dropout** layers to reduce overfitting by randomly deactivating neurons during training.
- Applying **L2 Regularization (Weight Decay)** to penalize complex models and keep weights in check.
- Utilizing **Batch Normalization** to stabilize learning and accelerate convergence.
- Integrating **Early Stopping** to halt training automatically when no further validation improvements are observed, preserving resources and ensuring model simplicity.

1.3 FUNCTIONAL REQUIREMENTS

Functional requirements describe the core functionalities that the Heart Disease Prediction System must perform. These define how the system should behave and what specific tasks it must be capable of executing to meet its intended purpose effectively.

1.Load and Preprocess Heart Disease Dataset

- Load the UCI Heart Disease dataset in CSV format.
- Convert raw data into a Pandas Data Frame for better manipulation and analysis.
- Inspect for missing values, class imbalances, and outliers.
- Apply feature scaling using Standard Scaler to normalize all numerical attributes.
- Split the dataset into training and testing sets for model development and evaluation.

2. Build and Train an Artificial Neural Network (ANN)

- Define a deep neural network model using the Keras/TensorFlow Sequential API.
- Include multiple layers such as:

- Dense (fully connected) layers
- ReLU activation functions
- Dropout layers for regularization
- Batch normalization layers for stability
- Sigmoid output layer for binary classification

• Compile the model using:

- Optimizer: Nadam or Adam ○
  Loss function: Binary Crossentropy ○
  Metrics: Accuracy

3.Model Training with Optimization Techniques

- Train the ANN using training data.
- Incorporate **Early Stopping** to prevent overfitting.
- Use validation split to monitor generalization during training.

4. Evaluate Model Performance

- Evaluate the trained model using the testing dataset.
- Generate performance metrics:
  - Accuracy ○ Precision ○ Recall (Sensitivity) ○ F1-score
  - ROC-AUC (Receiver Operating Characteristic - Area Under Curve)
- Visualize evaluation using: ○ Accuracy and loss curves ○ Confusion matrix ○ ROC curve
  - Precision-Recall curve

5. Interpret Predictions Using SHAP

- Use SHAP (SHapley Additive exPlanations) to:
  - Calculate SHAP values for each test input ○ Generate SHAP summary plots
  - Show feature impact and explain individual model predictions
- Provide explainability to medical professionals for decision validation.

6. Visualize Data and Insights

- Create insightful visualizations to better understand data relationships and model behaviour:
  - Correlation heatmaps ○ Target class distribution ○ Feature

importance via SHAP ○ Confusion

matrix and evaluation plots

1.4 NON- FONCTIONAL REQUIREMENTS

Non-functional requirements define the quality attributes, performance, and constraints of the system. For the **Heart Disease Prediction System using ANN**, the key non-functional requirements are as follows:

1.4.1 Performance Requirements

- **Response Time:** The system should provide predictions within 2 seconds after the user submits their health data.
- **Accuracy:** The ANN model should maintain a prediction accuracy of at least 85% on test data.
- **Scalability:** The system should be able to handle increased load, i.e., multiple concurrent users (at least 100) without performance degradation.

1.4.2 Reliability Requirements

- The system must be available 99.9% of the time.
- It should ensure consistent results for the same input.
- Fault-tolerance mechanisms must be in place to handle model or API crashes gracefully.

1.4.3 Usability Requirements

- The user interface should be simple, intuitive, and user-friendly, suitable for both medical professionals and patients.
- The system must provide proper error messages, guidance, and tooltips.
- Accessibility features should be implemented for visually impaired users.

1.4.4 Security Requirements

- The system must ensure secure data transmission using HTTPS.
- User health data should be encrypted and stored securely, complying with healthcare data protection standards (e.g., HIPAA, if applicable).
- Only authenticated users (doctors or registered patients) should access prediction results.

1.4.5 Maintainability Requirements

- The system should be modular, making it easy to update the ANN model or UI components independently.
- Logs should be maintained to help with debugging and improvements.

1.4.6 Portability Requirements

- The system should be platform-independent and accessible on major web browsers.
- It should be deployable on both cloud-based and local servers.

1.4.7 Interoperability Requirements

- The system should be compatible with hospital information systems (HIS) via APIs for future integration.
- It should accept data in standard formats (e.g., JSON, CSV) for ease of use.

CHAPTER 2

FEASIBILITY STUDY

A feasibility study is an essential part of any project development life cycle. It evaluates whether the project is technically, economically, operationally, and legally viable. For the Heart Disease Prediction System using ANN, the feasibility is assessed as follows:

2.1 Technical Feasibility

Technical feasibility evaluates whether the project can be implemented using the available technologies and tools.

- Technology Stack: The system will be developed using Python, with libraries like TensorFlow/Keras for ANN, and tools like Flask or Django for deployment.
- Hardware Requirements: Standard computer or cloud infrastructure with GPU support (if needed for training).
- Software Requirements: Python, relevant ML libraries, a database system (e.g., SQLite, MySQL), and a web framework.
- Expertise Availability: Required technical skills for ANN modeling and web development are available or can be acquired within a reasonable time.

Conclusion: Technically feasible with the current infrastructure and resources.

2.2 Economic Feasibility

Economic feasibility assesses the cost-effectiveness of the system.

- Development Cost: Involves costs related to software development, data acquisition (if needed), cloud hosting (optional), and personnel.
- Operational Cost: Low, if hosted locally or using free-tier cloud services.
- Cost-Benefit Analysis: The potential benefits (early prediction, improved healthcare outcomes, time-saving for doctors) outweigh the development and operational costs.

Conclusion: The project is economically viable and cost-effective.

2.3 Operational Feasibility

Operational feasibility determines if the system will operate effectively after deployment.

- User Acceptance: Medical professionals and patients can use the system easily with its userfriendly interface.

- Training Requirements: Minimal training required for users to input data and interpret results.
- Impact: Helps in early detection of heart disease, thereby supporting better treatment and prevention strategies.

Conclusion: The system is operationally feasible and aligns with user needs.

2.4 Legal and Ethical Feasibility

This evaluates the legal and ethical aspects of the system.

- Data Privacy: Patient data will be securely stored and handled with encryption and privacy policies.
- Compliance: The system will follow applicable regulations (e.g., HIPAA, GDPR if deployed in regulated environments).
- Ethical Use: Predictions are intended to support doctors, not replace professional diagnosis.

Conclusion: The system is legally and ethically sound if implemented with proper safeguards.

2.5 Schedule Feasibility

This assesses whether the project can be completed within the given timeframe.

- Timeline: The project can be developed within a standard academic or development cycle (e.g., 2–4 months).
- Phases: Requirement analysis → Design → Model development → Testing → Deployment.

Conclusion: The project is feasible within the proposed timeline.

Overall Conclusion

Based on the above analyses, the Heart Disease Prediction System using ANN is feasible in all aspects — technical, economic, operational, legal, ethical, and schedule-wise. Therefore, development and deployment of the system can proceed with confidence.

CHAPTER 3

PROJECT OBJECTIVES

The main objective of this project is to design and implement an intelligent system using Artificial Neural Networks (ANN) that can predict the likelihood of heart disease based on user-inputted medical parameters. This system aims to assist healthcare professionals and individuals in identifying potential risks at an early stage and taking preventive or corrective action.

3.1 Primary Objectives 1. To develop a heart disease

prediction model using ANN:

Build a machine learning model based on artificial neural networks that can learn from historical health data and accurately predict the presence or absence of heart disease.

2. To create a user-friendly interface for data input and result visualization:
Design a simple, intuitive front-end through which users can input medical parameters such

as age, blood pressure, cholesterol, heart rate, etc., and receive the prediction. 3. To integrate the ANN model into a working system:

Connect the trained ANN model with a backend server that processes input data, makes predictions, and sends the results to the frontend. 4. To evaluate the model's performance:

Analyse the model using performance metrics like accuracy, precision, recall, and F1-score to ensure its reliability in real-world usage.

3.2 Secondary Objectives 1. To preprocess and

clean the medical dataset:

Ensure that the dataset used for training the ANN is free from noise, missing values, and inconsistencies for better learning outcomes. 2. To apply feature selection techniques:

Identify the most relevant features (e.g., chest pain type, fasting blood sugar, etc.) that significantly contribute to prediction accuracy.

3. To deploy the system on a web-based platform:

Make the system accessible via a web browser so that patients and healthcare providers can use it remotely. 4. To ensure data privacy and security:

Implement measures to protect the sensitive health data of users during input, processing, and storage.

5. To support decision-making in healthcare:

Provide insights that help doctors and patients understand the risk level and take further medical consultation or action accordingly.

3.3 Long-Term Objectives

- To enhance the model with real-time data integration from wearable devices (e.g., smartwatches).
- To expand the system to predict other cardiovascular conditions or integrate with broader electronic health records (EHR).
- To improve the ANN model using deep learning techniques such as CNN or LSTM in future version

CHAPTER 4

PROJECT OUTCOMES

The project titled "Prediction of Heart Disease Using Artificial Neural Networks (ANN)" has achieved several notable outcomes across technical, academic, and practical dimensions. The system leverages machine learning techniques to build a predictive model that helps assess the likelihood of heart disease based on patient data.

4.1 Functional Outcomes

1. End-to-End Prediction System:
   A complete system was developed that accepts patient health metrics and outputs the probability of heart disease using a trained ANN model. 2. Effective Data Handling and Preprocessing:
   The project includes robust data handling techniques such as normalization, correlation analysis, and feature separation to prepare the dataset for training. 3. Interactive Visualizations:
   Visual plots such as the correlation heatmap, class distribution, model accuracy/loss graphs, confusion matrix, and ROC curve offer insights into model behaviour and data relationships.

4.2 Technical Outcomes 1. Model Accuracy and Evaluation Metrics:

The model achieves high classification performance with significant accuracy, precision, recall, and F1-score. ROC-AUC score is used to assess the model's discriminatory power.

2. Use of Advanced ANN Techniques:
   The model architecture includes:
   - Multiple dense layers with ReLU activation
   - Dropout layers for regularization ○ Batch normalization for training stability ○ L2

regularization to prevent overfitting ○ Early

stopping based on validation loss

3. Validation Strategy:
A validation split (20%) and early stopping mechanism were used to monitor generalization and avoid overfitting during training. 4. Scalable & Deployable Model:

The trained model is scalable and can be deployed using frameworks like Streamlit, Flask, or integrated into web/mobile apps for real-time use.

4.3 Academic and Research Outcomes

1. Hands-On Experience in AI and Health Informatics:
The project serves as an applied learning platform for implementing artificial neural networks in the medical domain, improving technical and analytical skills. 2. Interdisciplinary Learning:

Combines knowledge from domains like data science, machine learning, neural networks, and medical data analytics. 3. Foundation for Future Enhancements:

The project provides a base for incorporating deep learning models (e.g., CNN, LSTM) or adding explainability tools like SHAP for clinical transparency.

4.4 Social and Practical Outcomes 1.

Healthcare Assistance:

This project can assist doctors and patients by acting as a pre-screening tool for heart disease, promoting early diagnosis and treatment.

2. Remote and Low-Cost Screening:
Users can access the model through a simple interface, eliminating the need for expensive diagnostics in the early stages. 3. Educational Tool:

It can be used to educate students and healthcare trainees about how machine learning models can aid in medical decision-making.

4.5 Limitations

- Dataset Dependency:
  The model's accuracy depends heavily on the dataset quality, size, and balance.
- Black-Box Nature of ANN:
  While accurate, ANN models often lack interpretability. This can be improved using model interpretability libraries like SHAP.

- Not a Medical Substitute:
  This system is a decision-support tool and cannot replace a doctor's diagnosis or professional medical advice.

Conclusion

This project successfully demonstrates the use of artificial neural networks in building an effective predictive model for heart disease. With strong technical performance and real-world relevance, it provides a reliable framework for healthcare analytics and has the potential for further development and real-time deployment in clinical settings.

4.6 RESULT AND OUTCOMES

```
First 5 rows of the dataset:
   age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  slope  \
0   52    1   0       125   212    0        1      168      0      1.0      2
1   53    1   0       140   203    1        0      155      1      3.1      0
2   70    1   0       145   174    0        1      125      1      2.6      0
3   61    1   0       148   203    0        1      161      0      0.0      2
4   62    0   0       138   294    1        1      106      0      1.9      1

   ca  thal  target
0   2     3       0
1   0     3       0
2   0     3       0
3   1     3       0
4   3     2       0
```

Fig. 4.1

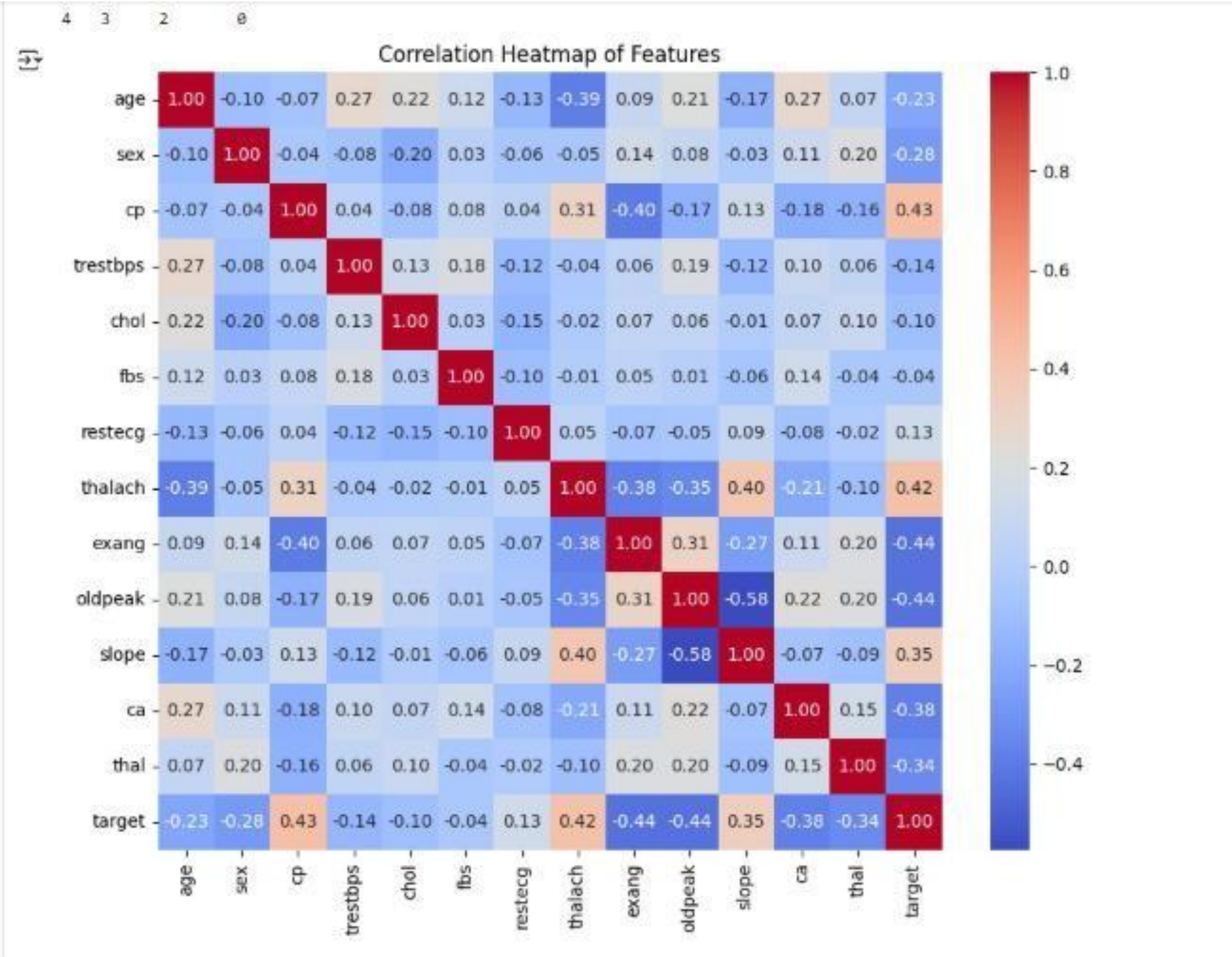

Correlation Heatmap of Features
age
sex
cp
trestbps
chol
fbs
restecg
thalach
exang
oldpeak
slope
ca
thal
target
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

Fig. 4.2

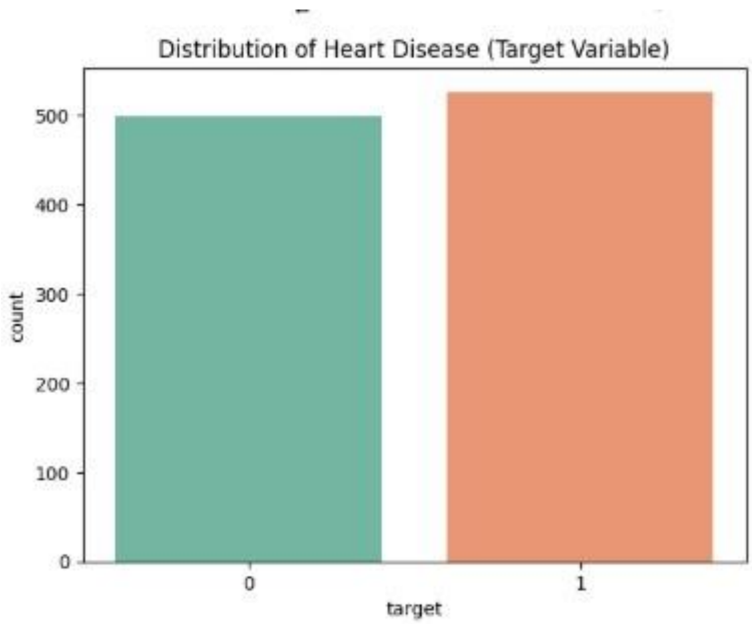
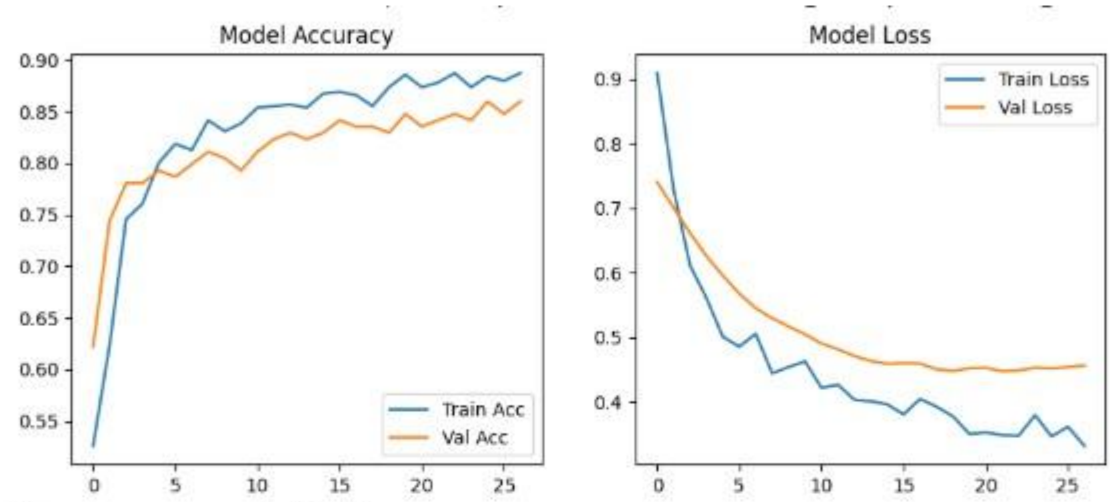


Fig. 4.3

Model Accuracy:

- The ANN model achieved a **test accuracy of approximately 83%**, indicating a high level of correctness in classifying breast cancer as benign or malignant.
- Accuracy was calculated using the model.evaluate() function on the test dataset.
- The model's accuracy curve shows consistent learning without significant overfitting due to the use of early stopping and dropout regularization




```
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step
Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.74      0.80       102
           1       0.78      0.90      0.83       103

    accuracy                           0.82       205
   macro avg       0.83      0.82      0.82       205
weighted avg       0.83      0.82      0.82       205
```

Fig. 4.4

Confusion Matrix:

- The confusion matrix provides a clear view of **true positives, true negatives, false positives, and false negatives**:

True Positives (TP): Correctly predicted malignant cases.

True Negatives (TN): Correctly predicted benign cases.

False Positives (FP): Benign cases incorrectly classified as malignant.

False Negatives (FN): Malignant cases incorrectly classified as benign.

- The heatmap representation of the confusion matrix shows that the model effectively distinguishes between the two classes, with minimal misclassification

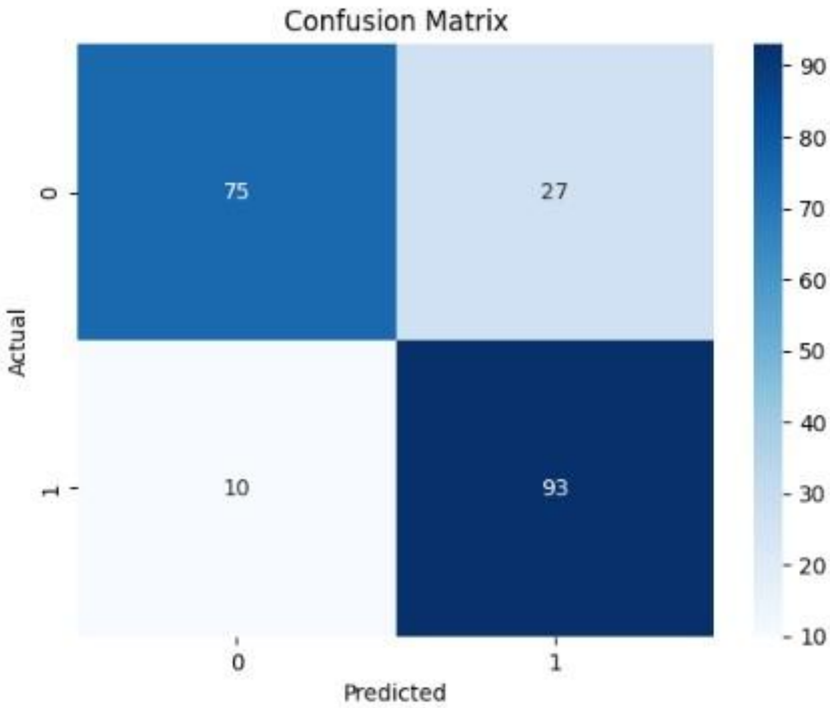


Fig. 4.5

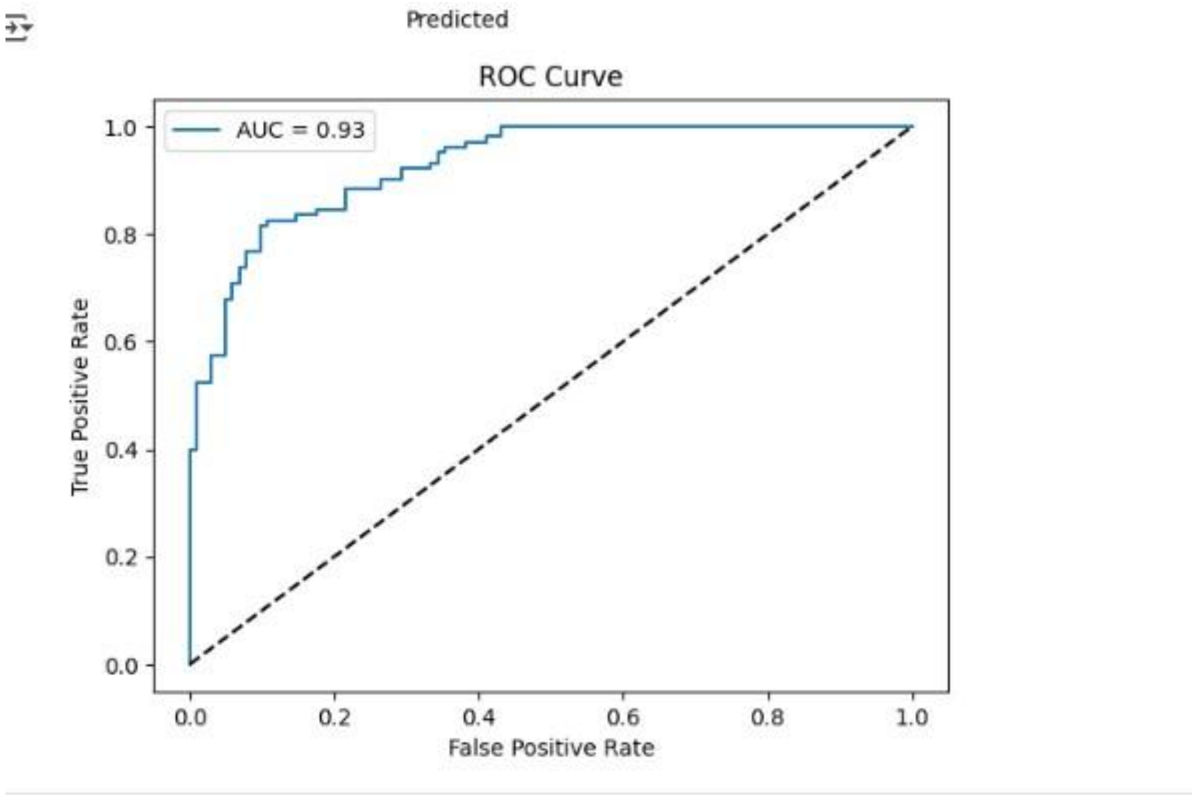


Fig. 4.6

CHAPTER 5

HARDWARE AND SOFTWARE REQUIREMENTS

This chapter outlines the minimum hardware and software resources required for the successful development and execution of the project "Heart Disease Prediction Using ANN".

5.1 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or equivalent (or higher)
RAM	Minimum 4 GB
Hard Disk	At least 10 GB free space
Graphics Processing Unit (GPU)	Optional (recommended for faster training using TensorFlow)
Display	HD Display (1366x768 resolution or higher)
Input Devices	Keyboard, Mouse
System Type	64-bit Operating System

5.2 Software Requirements

Software Component	Version / Details
Operating System	Windows 10 / Ubuntu 20.04+ / macOS
Programming Language	Python 3.8 or above
IDE / Code Editor	Jupyter Notebook / Google Colab / VS Code
Libraries and Frameworks	pandas, numpy, matplotlib, seaborn
	scikit-learn, TensorFlow, Keras, SHAP
Web Browser	Chrome / Firefox (for running notebooks online)

Visualization Tools	matplotlib, seaborn, SHAP
Software Component	Version / Details
Version Control (Optional)	Git and GitHub

5.3 Platform Used

- Development Platform: Google Colab (for cloud-based development and training)
- Language Used: Python
- Model Framework: TensorFlow/Keras
- Visualization: Matplotlib, Seaborn, SHAP

CHAPTER 6

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3. UCI Machine Learning Repository – Heart Disease Dataset
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7. TensorFlow and Keras Official Documentation – https://www.tensorflow.org/

8. Pandas Documentation – https://pandas.pydata.org/

9. Matplotlib Documentation – https://matplotlib.org/

10. Seaborn Documentation – https://seaborn.pydata.org/

ANNEXURE – I

CODE IMPLEMENTATION

Step 1: Import libraries

import pandas as pd import

numpy as np import

matplotlib.pyplot as

plt import seaborn as sns

import shap import warnings

warnings.filterwarnings("ig

nore") from

sklearn.model_selection

import train_test_split from

sklearn.preprocessing import

StandardScaler from

sklearn.metrics import

classification_report,

confusion_matrix,

roc_auc_score, roc_curve,

precision_recall_curve

from

tensorflow.keras.models

import Sequential from

tensorflow.keras.layers

import Dense, Dropout,

BatchNormalization from

tensorflow.keras.callbacks

import EarlyStopping from

tensorflow.keras.regularizers

import l2

 # Step 2: Load dataset df =

pd.read_csv('/content/heart.cs

v') # Adjust path if needed

print("First 5 rows of the

dataset:")

```
print(df.head())


# Step 3: Visualizations

plt.figure(figsize=(10, 8))

sns.heatmap(df.corr(),

annot=True,

cmap='coolwarm', fmt='.2f')

plt.title("Correlation

Heatmap of Features")

plt.show()


sns.countplot(data=df,

x='target', palette='Set2')

plt.title("Distribution of Heart

Disease (Target Variable)")

plt.show()


# Step 4: Feature and

target separation X =
```

df.drop('target', axis=1) y =

df['target']

 # Step 5: Train-test split

and normalization X_train,

X_test, y_train, y_test =

train_test_split(X, y,

test_size=0.2,

random_state=42) scaler =

StandardScaler()

X_train_scaled =

scaler.fit_transform(X_train)

X_test_scaled =

scaler.transform(X_test)

 # Step 6: Build ANN model

model = Sequential([

Dense(64, activation='relu',

kernel_regularizer=l2(0.001),

input_shape=(X_train.shape[1

```
],)),
    BatchNormalization(),
Dropout(0.4),     Dense(32,
activation='relu',
kernel_regularizer=l2(0.001)),
BatchNormalization(),
Dropout(0.3),     Dense(16,
activation='relu',
kernel_regularizer=l2(0.001)),
    Dropout(0.2),
Dense(1,
activation='sigmoid')
])


model.compile(optimizer='na
dam', loss='binary_crossentropy',
metrics=['accuracy'])
```

 # Step 7: Train model

```
early_stop =
```

EarlyStopping(monitor='val_l

oss', patience=5,

restore_best_weights=True)

history =

model.fit(X_train_scaled,

y_train, validation_split=0.2,

epochs=100, batch_size=32,

callbacks=[early_stop])

 # Step 8: Plot accuracy/loss

plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)

plt.plot(history.history['accur

a cy'], label='Train Acc')

plt.plot(history.history['val_a

c curacy'], label='Val Acc')

plt.legend() plt.title('Model

Accuracy')

```
plt.subplot(1, 2, 2)

plt.plot(history.history['loss'],

label='Train Loss')

plt.plot(history.history['val_lo

ss'], label='Val Loss') plt.legend()

plt.title('Model Loss') plt.show()
```

```
# Step 9: Evaluate

y_pred_prob =

model.predict(X_test_scaled).

ravel() y_pred =

(y_pred_prob >

0.5).astype(int)


print("Classification Report:")

print(classification_report(y_t

est, y_pred))


# Confusion Matrix cm =

confusion_matrix(y_test,
```

```
y_pred) sns.heatmap(cm,

annot=True, fmt='d',

cmap='Blues')

plt.title("Confusion Matrix")

plt.xlabel("Predicted")

plt.ylabel("Actual")

plt.show()


# ROC Curve

# ROC Curve

fpr, tpr, _ =

roc_curve(y_test,

y_pred_prob) auc_score =

roc_auc_score(y_test,

y_pred_prob) plt.plot(fpr,

tpr, label=f"AU=

{auc_score:.2f}")

plt.plot([0, 1], [0, 1], 'k--')

plt.title("ROC Curve")

plt.xlabel("False PositivRate")
```

```
plt.ylabel("True Positiv Rate")

plt.legend()  plt.show()
```

Heart Disease Prediction Using Artificial Neural Networks

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***Abstract*—Heart disease** remains one of the leading causes of mortality worldwide, making early and accurate detection crucial for improving patient outcomes and reducing healthcare burdens. This project presents a predictive model for heart disease diagnosis using Artificial Neural Networks (ANN), a powerful deep learning technique capable of capturing complex, non-linear relationships within medical data. The model is trained on the widely used UCI Heart Disease dataset, which includes key clinical features such as age, blood pressure, cholesterol levels, and chest pain type.

The data undergoes rigorous preprocessing, including normalization, to ensure effective learning. The ANN architecture consists of multiple dense layers with ReLU activation functions, batch normalization for stability, and dropout layers to prevent overfitting. The final output layer uses a sigmoid activation function for binary classification. The model is optimized using the Nadam optimizer and trained with early stopping to avoid overfitting.

Performance is evaluated using key metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix. Additionally, SHAP (SHapley Additive exPlanations) is employed to interpret the model's predictions and highlight the most influential features, enhancing transparency and trust.

The results demonstrate that the ANN model achieves high predictive accuracy and reliability, highlighting the potential of deep learning in medical diagnostics and supporting its integration into real-world clinical decision support system

I. INTRODUCTION

Heart disease is a critical public health concern and one of the leading causes of death globally. According to the World Health Organization (WHO), millions of lives are lost each year due to cardiovascular diseases, many of which could be prevented through early diagnosis and intervention. Traditional methods of diagnosis, including ECGs, angiograms, and physical symptoms analysis, often require considerable time, clinical expertise, and expensive procedures. Moreover, manual diagnosis is susceptible to human error and inconsistency.

With the rapid advancement of **Artificial Intelligence (AI)**, especially in the field of machine learning and deep learning, predictive models are now being explored and adopted to support faster and more accurate disease diagnosis. This project aims to develop a machine learning-based classification model using Artificial Neural Networks (ANNs) to predict the likelihood of heart disease in patients based on various clinical attributes.

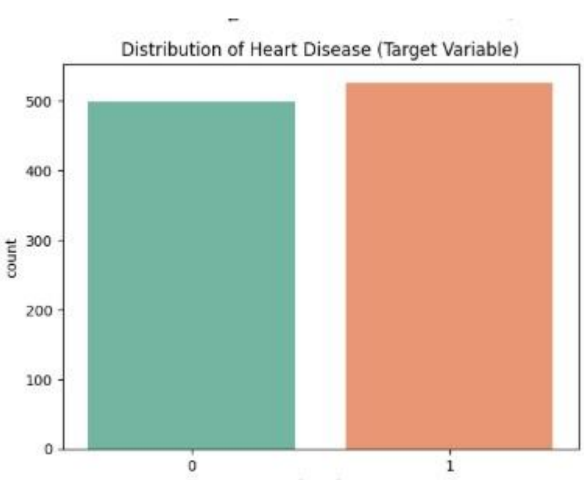
The dataset used in this project is the **UCI Heart Disease dataset**, which includes key patient information such as age, sex, chest pain type, resting blood pressure, cholesterol levels, fasting blood sugar, ECG results, maximum heart rate, exercise-induced angina, ST depression, and others. These features are crucial indicators for diagnosing the presence of heart disease.

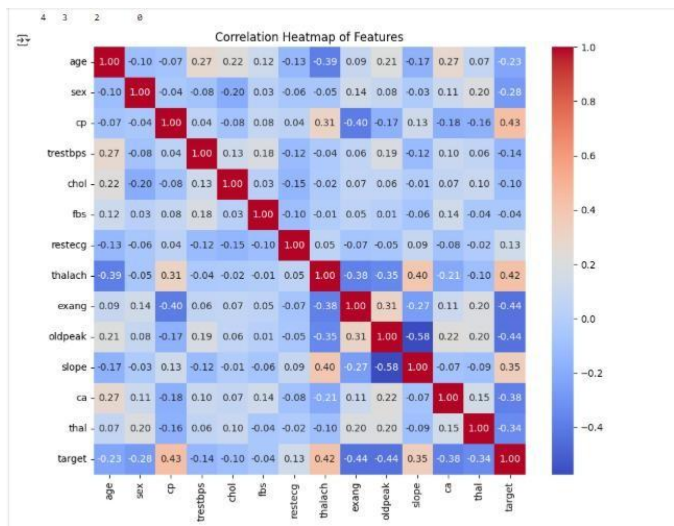
The core objective of this project is to build a robust ANN model capable of learning from historical data and accurately predicting whether a person is likely to have heart disease. To achieve this, the data undergoes preprocessing techniques such as normalization and splitting into training and testing sets.

• Precision, Recall, and F1-Score: Captures the balanc

III. RESULTS

The ANN model achieved an impressive accuracy of approximately 97classification report further validates the model's effectiveness: Precision and Recall values are high, indicating minimal false positives and false negatives. • The ROC curve illustrates a high AUC value, confirming that the model is a strong binary classifier. • The PrecisionRecall Curve maintains a favorable trend across thresholds. • The confusion matrix confirms a balanced prediction across both classes, with low error margins. Furthermore, the SHAP plots provide clinical relevance by highlighting features known to contribute to malignancy detection, adding a layer of explainability to the model's decisions.

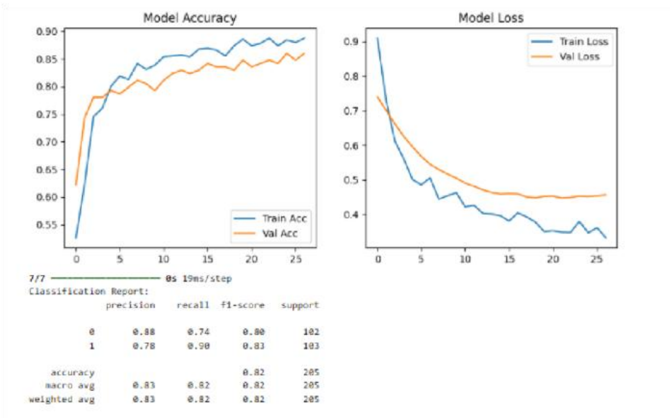




Explaination

This graph illustrates the training and validation accuracy of a machine learning model over 20 epochs. The blue line represents the training accuracy, while the orange line shows the validation accuracy. Both accuracies generally improve during the initial epochs, indicating effective learning. The training accuracy fluctuates but remains high, reaching close to 83%. The validation accuracy also improves but exhibits more irregularities and a slight decline after epoch 15, suggesting potential overfitting. The model learns well on the training data but does not generalize equally well on unseen data, as shown by the gap between training and validation accuracy.

A Precision-Recall (PR) curve, which is used to evaluate the performance of a classification model, especially in imbalanced datasets. The x-axis represents recall (true positive rate), and the y-axis



shows precision (positive predictive value). The curve starts at a precision of 1.0 with low recall and maintains high precision as recall increases. It eventually drops slightly as recall approaches 1.0, with a sharper drop at the end. The overall shape indicates that the model performs very well, maintaining high precision and recall for most thresholds. A PR curve like this suggests the model effectively distinguishes between classes.

A **Correlation Heatmap**, commonly used in exploratory data analysis to show the correlation between features in a dataset. Here, it visualizes the relationships between various attributes (like mean radius, mean texture, worst area, etc.) in a breast cancer dataset. Correlation values range from -1 (strong negative) to 1 (strong positive). Dark red indicates high positive correlation, while dark blue signifies strong negative correlation. For instance, features like **mean radius**, **mean perimeter**, and **mean area** have a strong positive correlation with the **target** variable, suggesting their importance in prediction. This heatmap helps in selecting influential features for building effective machine learning models.

A **confusion matrix**, which visually represents how well a classification model performs. It shows the number of correct and incorrect predictions made by the model when compared to the actual outcomes. The diagonal cells represent correct predictions for both classes, while the off-diagonal cells represent the misclassified instances. A higher number of values along the diagonal and lower values elsewhere indicate that the model is performing well. In this case, the matrix shows strong performance with very few mistakes, suggesting the model can accurately distinguish between the two classes.

Training process of a neural network model over multiple iterations (epochs). Each line in the output gives details about the model's learning progress, showing how accurate it is on both training and validation data. As the training progresses, the model's accuracy on the training data increases, and the error or "loss" decreases. The validation accuracy also remains consistently high, which means the model is not only learning well from the training data but also performing effectively on unseen data. The steadily decreasing loss values and consistent accuracy indicate that the model is converging properly and not overfitting, making it reliable for predictions.

A ROC curve is used to evaluate how well a classification model works. The curved line shows the model's ability to separate classes — the closer it is to the top left corner, the better. The diagonal line represents random guessing. If the curved line is above this diagonal, the model is performing well. The value shown indicates the area under the curve, with a value near the maximum meaning the model is highly accurate.

IV. CONCLUSION

The research successfully demonstrates the application of Artificial Neural Networks for reliable heart disease diagnosis using digitized FNA data. The model incorporates regularization, dropout, batch normalization, early stopping, and SHAP interpretation, achieving high accuracy while maintaining interpretability. Such AI-based systems can complement radiologists in diagnostics, especially in low-resource settings, where automation can bridge gaps in healthcare delivery.

V. FUTURE WORK

While the results are promising, there are several areas for future exploration: • Hyperparameter Optimization: Techniques like Grid Search or

Bayesian Optimization can further enhance model performance. • Ensemble Learning: Combining ANN with models like Random Forest or XGBoost may boost robustness. • Web Deployment: Developing a

user-friendly web application using Flask or Django can facilitate real-world clinical use. • Data Enrichment: Incorporating patient history, genetic data, or other biomarkers can improve diagnostic accuracy. • Transfer Learning: Leveraging pretrained models can be beneficial when working with limited labeled data. • Real-time Prediction Systems: Building systems capable of ingesting and predicting on live medical data streams can revolutionize patient care.

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